

Aptamer Design Lab – Technical Case Study

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- 1. Approach and Workflow:** I installed AlphaFold3 and Boltz-2 natively via uv on the Andromeda cluster after confirming Singularity was non-functional due to permission restrictions. I built a reproducible pipeline (GitHub, commit 80c59c3) that filters the UTexas Aptamer Database to 73 ssDNA/ssRNA aptamers with numeric Kd across 7 single-protein targets (≥ 7 clones each: IgE, HIV-1 Rev, VEGF, bFGF, Thrombin, Abrin, ABF1), corrects target sequences to their mature/processed UniProt form, and auto-generates AF3/Boltz-2 inputs. Claude and Cursor were used throughout for debugging, architecture investigation, and automation.
- 2. Technical Challenges:** Singularity was broken (missing fakeroot mapping, then setuid errors), resolved via native install. Boltz-2 rejects DNA/RNA as an affinity binder by validation only, not architecture — the tokenizer assigns affinity by chain ID, not molecule type — so I patched the schema, `parse_polymer()`, and a required `mw` field. Wiring AF3 coordinates into Boltz's affinity head (Task 6) initially matched only 59% of heavy atoms due to a 0- vs. 1-index residue offset; fixed to 100%.
- 3. Predicted vs. Experimental Affinity Analysis:** Spearman correlation was used, with a sign convention worth noting: agreement with Kd is a positive rho for `affinity_pred_value`, but negative rho for `affinity_probability_binary` and AF3 ipTM. Boltz-2 pooled correlation (n=73) was rho = -0.16 (p = 0.18) without MSA and rho = -0.18 (p = 0.13) with MSA (mean per-target rho: +0.17 and -0.03, respectively — MSA did not help). AF3 ipTM pooled at rho = +0.25 (p = 0.04), the wrong direction. The only significant, correctly directed per-target result was Abrin/AF3-ipTM (rho = -0.79, p = 0.02, n = 8). In the bonus task, wiring AF3 coordinates into Boltz's affinity head gave almost the same score as Boltz's own pose on the same aptamer (-0.531 vs -0.527), suggesting the head relies more on sequence embeddings than 3D coordinates. Leave-one-target-out fine-tuning worsened generalization (mean held-out rho: +0.16 zero-shot, -0.09 heads only, -0.30 full module), consistent with overfitting to the training targets.
- 4. Conclusions and Limitations:** Aside from Abrin ipTM (n = 8), no method showed significant, correctly directed correlation once evaluated per-target rather than pooled. Results are specific to this 73-aptamer/7-target subset: n per target is small (8-13), HIV-1 Rev has no Kd variation across its 12 clones, and pooled correlations largely reflect between-protein scale differences rather than true within-target ranking ability.
- 5. Next Steps:** Prioritize more clones with one well-measured protein over more targets, to isolate ranking ability from protein-identity confounds. The near-identical zero-shot scores under AF3 vs. Boltz-2 poses are worth probing directly (e.g., deliberately perturbed structures) before further fine-tuning investment.